PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning

Project Description:

PlantCare AI aims to develop an accurate and efficient model for classifying plant diseases by employing transfer learning techniques. Utilizing the New Plant Diseases Dataset containing over 87,000 annotated plant leaf images categorized into 38 distinct classes including various diseases affecting crops like Apple, Corn, Grape, Potato, Tomato, and others, the project leverages the pre-trained MobileNetV2 convolutional neural network (CNN) to expedite training and improve classification accuracy. Transfer learning allows the model to benefit from pre-existing knowledge of image features, significantly enhancing its performance and reducing computational costs. This approach provides a reliable and scalable tool for farmers, agricultural professionals, and gardeners, ensuring precise and efficient plant disease identification.

Application Scenarios

Scenario 1: Automated Agricultural Monitoring Systems

Integrating PlantCare AI into automated monitoring systems on farms can revolutionize crop health management. By using transfer learning, the system quickly adapts to the specifics of plant disease classification, capturing images of plant leaves, classifying diseases in real-time, and generating detailed reports. This automation reduces the manual workload on agricultural experts, speeds up diagnostic processes, and ensures high accuracy in results, ultimately improving crop yields and reducing losses due to disease.

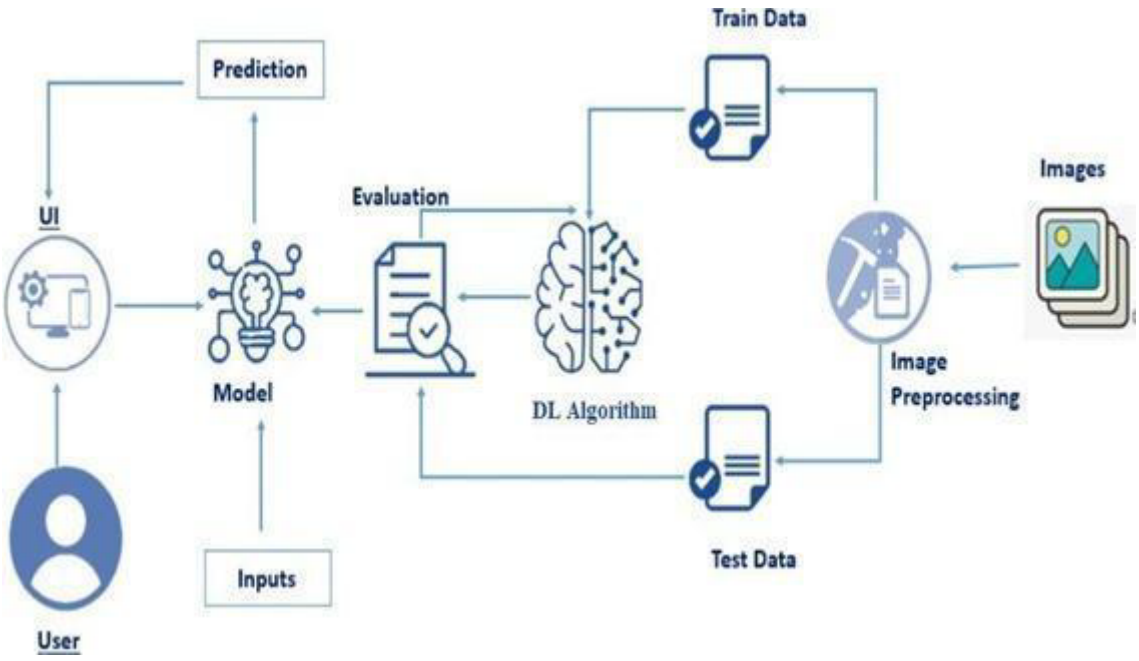
Scenario 2: Mobile Applications for Home Gardeners

PlantCare AI can be deployed as a mobile application to help home gardeners and hobbyists identify plant diseases. With transfer learning, the model's ability to accurately classify diseases from diverse plant images is improved, allowing users to simply take photos of affected leaves for instant analysis. This enables timely intervention without the need for expert consultation, making plant care more accessible and helping maintain healthy gardens.

Scenario 3: Educational Tools for Agricultural Training

PlantCare AI's transfer learning-based classification model can be integrated into educational platforms for agricultural training. By incorporating this advanced technology into interactive learning tools, students and agricultural technicians can upload and analyze plant images to receive instant feedback. This hands-on learning experience enhances their understanding of plant pathology and disease identification, providing practical skills crucial for modern agricultural practices.

Technical Diagram:



Prerequisites:

To complete this project, you must require the following software, concepts, and packages

- **Anaconda Navigator:**
  - o Refer to the link below to download Anaconda Navigator
- **Python packages:**
  - o Open anaconda prompt as administrator
  - o Type “pip install numpy” and click enter.
  - o Type “pip install pandas” and click enter.
  - o Type “pip install scikit-learn” and click enter.
  - o Type ”pip install matplotlib” and click enter.

o Type ”pip install scipy” and click enter.

o Type ”pip install seaborn” and click enter.

o Type ”pip install tenserflow” and click enter.

o Type “pip install Flask” and click enter.

Prior Knowledge:

You must have prior knowledge of the following topics to complete this project.

● **DL Concepts**

o**Neural Networks:**

https://www.analyticsvidhya.com/blog/2024/05/ann-vs-cnn-vs-rnn/?utm_source=chatgpt.com

Deep Learning Frameworks:
https://www.analyticsvidhya.com/blog/2019/03/deep-learning-frameworks-comparison/?utm_source=chatgpt.com

Transfer Learning:
https://www.datacamp.com/blog/what-is-transfer-learning-in-ai-an-introductory-guide?utm_source=chatgpt.com

Mobilenet_v2:
https://www.analyticsvidhya.com/blog/2023/12/what-is-mobilenetv2/?utm_source=chatgpt.com

Convolutional Neural Networks (CNNs):
https://www.analyticsvidhya.com/blog/2022/01/introduction-to-neural-networks/?utm_source=chatgpt.com

Overfitting and Regularization:
https://www.analyticsvidhya.com/blog/2018/11/neural-networks-hyperparameter-tuning-regularization-deeplearning/?utm_source=chatgpt.com

Optimizers:
https://www.analyticsvidhya.com/blog/2023/09/what-is-adam-optimizer/?utm_source=chatgpt.com

● Flask Basics: https://www.youtube.com/watch?v=lj4I_CvBnt0

Project Objectives

By the end of this project, you will: - Understand fundamental concepts and techniques used for Deep Learning in image classification - Gain comprehensive knowledge of agricultural disease datasets - Master data preprocessing and augmentation techniques for plant disease images - Learn to implement transfer learning using MobileNetV2 - Develop skills in model evaluation and performance optimization - Create a full-stack web application integrating the trained model - Deploy an AI-powered plant disease detection system

Project Flow:

The user interacts with the web interface following this flow:

1 .User Interface Interaction:

User accesses the web application and navigates to the upload page

2. Image Selection:

User selects or captures an image of a plant leaf

3. Image Processing:

The uploaded image is preprocessed and resized to match model requirements

4. Model Prediction:

The integrated MobileNetV2 model analyzes the image

5. Result Display:

Prediction results with disease classification and recommendations are displayed

Complete Project Activities:

1. Data Collection:

Download and organize the New Plant Diseases Dataset

2. Data Preprocessing

- Data exploration and visualization
- Image augmentation (already applied in dataset)
- Dataset splitting into training and validation sets

3. Model Building

- Import necessary libraries
- Load pre-trained MobileNetV2 model
- Fine-tune the model architecture
- Configure model compilation
- Set up training callbacks

4.Model Training

- Train the model with data generators
- Monitor training progress
- Implement early stopping and learning rate reduction

5. Model Evaluation

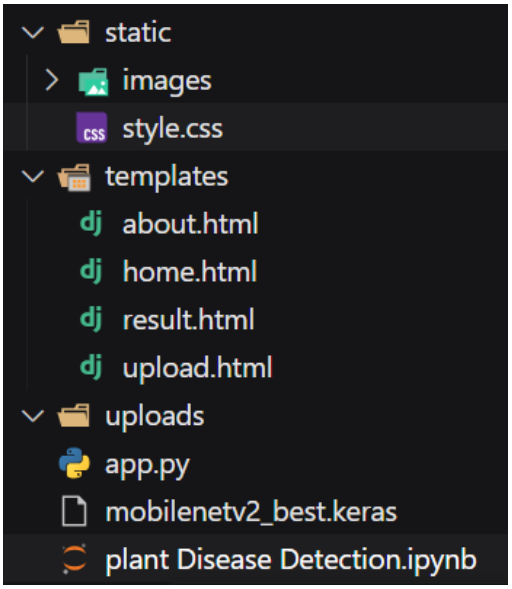
- Evaluate model performance on validation set
- Visualize training metrics
- Save the best performing model

6.Application Development

- Create HTML templates for web interface
- Develop Flask backend
- Integrate model with web application
- Test end-to-end functionality

Project Structure:

Create the Project folder which contains files as shown below



Project Structure Explanation

- static/

  Contains static assets (CSS, JavaScript, uploaded images)

  - images/ – Stores user-uploaded images for display
  - style.css – Application styling

- templates/

  HTML templates for Flask application

  - home.html – Landing page
  - about.html – Information about the technology
  - upload.html – Image upload interface
  - result.html – Disease prediction results

- uploads/

  Temporary storage for file uploads

- app.py

  Flask application backend

- mobilenetv2_best.keras

  Trained model file

- plant Disease Detection.ipynb

  Jupyter notebook with model training code

Milestone 1: Data Collection & Data Preprocessing

Activity 1: Download the Dataset

Dataset Information:

The **New Plant Diseases Dataset** is a comprehensive collection designed for **plant pathology research** and image-based **disease classification**.

- **Total Images:** 87,000+ augmented RGB images
- **Image Format:** JPEG
- **Number of Classes:** 38 (including healthy and diseased categories)
- **Plant Types:** Apple, Blueberry, Cherry, Corn, Grape, Orange, Peach, Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, Tomato
- **Image Size:** 256×256 pixels *(resized to 224×224 for MobileNetV2)*

Dataset Source

[Kaggle – New Plant Diseases Dataset]

Dataset Structure

```
new-plant-diseases-dataset/
├── New Plant Diseases Dataset(Augmented)/
│   ├── train/
│   │   ├── Apple___Apple_scab/
│   │   ├── Apple___Black_rot/
│   │   ├── Apple___Cedar_apple_rust/
│   │   ├── Apple___healthy/
│   │   ├── ... (38 classes total)
│   │   └── Tomato___healthy/
│   └── valid/
│       ├── Apple___Apple_scab/
│       ├── ... (same 38 classes)
│       └── Tomato___healthy/
```

Activity 1.1: Setting Up the Environment

Step 1: Upload Dataset to Google Colab

First, we need to set up our Kaggle API credentials to download the dataset directly in Colab:

```
# Upload your kaggle.json file
from google.colab import files
files.upload()  # This will prompt you to upload kaggle.json
```

```
Choose Files kaggle.json
kaggle.json(application/json) - 70 bytes, last modified: 9/15/2025 - 100% done
Saving kaggle.json to kaggle.json
```

Activity 1.2: Importing Required Libraries

```
# Install required packages
!pip install -q kaggle "tensorflow>=2.17.0" gradio matplotlib

# Python imports & configuration
import os
import sys
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

Activity 1.3: Download and Extract Dataset

```
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# If dataset not downloaded yet, download & unzip (comment out if already present)
import os
if not os.path.exists("/content/new-plant-diseases-dataset"):
    !kaggle datasets download -d vipoooool/new-plant-diseases-dataset -p /content
    !unzip -q /content/new-plant-diseases-dataset.zip -d /content/new-plant-diseases-dataset

# Confirm dataset presence (small print)
!ls -lah /content/new-plant-diseases-dataset | sed -n '1,200p'
```

```
Dataset URL: https://www.kaggle.com/datasets/vipoooool/new-plant-diseases-dataset
License(s): copyright-authors
Downloading new-plant-diseases-dataset.zip to /content
 97% 2.61G/2.70G [00:27<00:02, 36.6MB/s]
100% 2.70G/2.70G [00:27<00:00, 107MB/s]
total 20K
drwxr-xr-x 5 root root 4.0K Oct 31 06:24 .
drwxr-xr-x 1 root root 4.0K Oct 31 06:23 ..
drwxr-xr-x 3 root root 4.0K Oct 31 06:23 new plant diseases dataset(augmented)
drwxr-xr-x 3 root root 4.0K Oct 31 06:23 New Plant Diseases Dataset(Augmented)
drwxr-xr-x 3 root root 4.0K Oct 31 06:24 test
```

Activity 2: Data Exploration and Visualization

Activity 2.1: Configure Dataset Paths

```
# User / experiment configuration
IMG_SIZE = (224, 224)
BATCH_SIZE = 32
EPOCHS = 5
N_LAST_LAYERS = 10   # Unfreeze last 10 layers for fine-tuning
SEED = 1337

# Dataset paths
train_dir = "/content/new-plant-diseases-dataset/new plant diseases dataset(augmented)/New Plant Diseases Dataset(Augmented)/train"
valid_dir = "/content/new-plant-diseases-dataset/new plant diseases dataset(augmented)/New Plant Diseases Dataset(Augmented)/valid"

# Verify paths exist
for p in [train_dir, valid_dir]:
    if not os.path.exists(p):
        print(f"ERROR: path not found: {p}")
        sys.exit(1)

print("train_dir:", train_dir)
print("valid_dir:", valid_dir)
```

```
train_dir: /content/new-plant-diseases-dataset/new plant diseases dataset(augmented)/New Plant Diseases Dataset(Augmented)/train
valid_dir: /content/new-plant-diseases-dataset/new plant diseases dataset(augmented)/New Plant Diseases Dataset(Augmented)/valid
```

Activity 2.2: Visualize Sample Images

```python
import random
from IPython.display import Image, display

# Function to display random samples from each class
def show_sample_images(base_path, num_classes=5, images_per_class=3):
    """Display random sample images from the dataset"""
    class_names = os.listdir(base_path)
    random_classes = random.sample(class_names, min(num_classes, len(class_names)))

    fig, axes = plt.subplots(num_classes, images_per_class,
                             figsize=(12, 3*num_classes))
    fig.suptitle('Sample Images from Dataset', fontsize=16)

    for i, class_name in enumerate(random_classes):
        class_path = os.path.join(base_path, class_name)
        images = os.listdir(class_path)
        random_images = random.sample(images, min(images_per_class, len(images)))

        for j, img_name in enumerate(random_images):
            img_path = os.path.join(class_path, img_name)
            img = plt.imread(img_path)
            axes[i, j].imshow(img)
            axes[i, j].axis('off')
            if j == 0:
                axes[i, j].set_title(class_name.replace('___', '\n'), fontsize=10)

    plt.tight_layout()
    plt.show()
```

Activity 2.3: Dataset Statistics

```python
# Count images in each split
def count_images(directory):
    """Count total images and images per class"""
    total_images = 0
    class_counts = {}

    for class_name in os.listdir(directory):
        class_path = os.path.join(directory, class_name)
        if os.path.isdir(class_path):
            num_images = len(os.listdir(class_path))
            class_counts[class_name] = num_images
            total_images += num_images

    return total_images, class_counts

# Get statistics
train_total, train_counts = count_images(train_dir)
valid_total, valid_counts = count_images(valid_dir)

print(f"Training images: {train_total}")
print(f"Validation images: {valid_total}")
print(f"Number of classes: {len(train_counts)}")
print(f"\nClass distribution (first 10):")
for i, (class_name, count) in enumerate(list(train_counts.items())[:10]):
    print(f"  {class_name}: {count} images")
```

```
Training images: 70295
Validation images: 17572
Number of classes: 38

Class distribution (first 10):
  Corn_(maize)___healthy: 1859 images
  Peach___healthy: 1728 images
  Soybean___healthy: 2022 images
  Tomato___Bacterial_spot: 1702 images
  Pepper,_bell___healthy: 1988 images
  Tomato___Late_blight: 1851 images
  Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot: 1642 images
  Apple___Apple_scab: 2016 images
  Blueberry___healthy: 1816 images
  Grape___Leaf_blight_(Isariopsis_Leaf_Spot): 1722 images
```

Activity 3: Data Preprocessing and Augmentation

Note: The dataset used in this project already contains augmented images. The original dataset was augmented using various techniques including rotation, flipping, and brightness adjustments. For this reason, we'll apply minimal additional augmentation during training.

Activity 3.1: Setup Data Generators

```
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input

# Training data generator with light augmentation
train_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    horizontal_flip=True,
    rotation_range=20,
    zoom_range=0.15,
    width_shift_range=0.1,
    height_shift_range=0.1,
    fill_mode='reflect'
)

# Validation data generator (no augmentation)
valid_datagen = ImageDataGenerator(preprocessing_function=preprocess_input)

# Create data generators
train_gen = train_datagen.flow_from_directory(
    train_dir,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    shuffle=True,
    seed=SEED
)

valid_gen = valid_datagen.flow_from_directory(
    valid_dir,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    shuffle=False
)
```

```
Found 70295 images belonging to 38 classes.
Found 17572 images belonging to 38 classes.
```

Activity 3.2: Extract Class Names

```
# Get class names and count
class_names = list(train_gen.class_indices.keys())
NUM_CLASSES = len(class_names)

print(f"Detected classes: {NUM_CLASSES}")
print("\nClass names:")
for i, name in enumerate(class_names):
    print(f"{i+1}. {name}")
```

```
Detected classes: 38

Class names:
1. Apple___Apple_scab
2. Apple___Black_rot
3. Apple___Cedar_apple_rust
4. Apple___healthy
5. Blueberry___healthy
6. Cherry_(including_sour)___Powdery_mildew
7. Cherry_(including_sour)___healthy
8. Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot
9. Corn_(maize)___Common_rust_
10. Corn_(maize)___Northern_Leaf_Blight
11. Corn_(maize)___healthy
12. Grape___Black_rot
13. Grape___Esca_(Black_Measles)
14. Grape___Leaf_blight_(Isariopsis_Leaf_Spot)
15. Grape___healthy
16. Orange___Haunglongbing_(Citrus_greening)
17. Peach___Bacterial_spot
18. Peach___healthy
19. Pepper,_bell___Bacterial_spot
20. Pepper,_bell___healthy
21. Potato___Early_blight
22. Potato___Late_blight
23. Potato___healthy
24. Raspberry___healthy
25. Soybean___healthy
26. Squash___Powdery_mildew
27. Strawberry___Leaf_scorch
28. Strawberry___healthy
29. Tomato___Bacterial_spot
30. Tomato___Early_blight
31. Tomato___Late_blight
32. Tomato___Leaf_Mold
33. Tomato___Septoria_leaf_spot
34. Tomato___Spider_mites Two-spotted_spider_mite
35. Tomato___Target_Spot
36. Tomato___Tomato_Yellow_Leaf_Curl_Virus
37. Tomato___Tomato_mosaic_virus
38. Tomato___healthy
```

Milestone 2: Model Building

Activity 1: Understanding Transfer Learning with MobileNetV2

What is Transfer Learning?

Transfer learning is a machine learning technique where a model developed for one task is reused as the starting point for a model on a second task. It's a popular approach in deep learning where pre-trained models are used as the starting point on computer vision and natural language processing tasks.

Why MobileNetV2?

MobileNetV2 is specifically designed for mobile and embedded vision applications. It features:

- Efficient Architecture: Uses depthwise separable convolutions
- Inverted Residuals: Linear bottleneck layers
- Lightweight: Fewer parameters compared to other architectures
- High Accuracy: Maintains competitive accuracy despite smaller size
- Pre-trained Weights: Trained on ImageNet with 1000 classes

Activity 2: Load Pre-trained MobileNetV2

```
from tensorflow.keras.applications import MobileNetV2

# Load pretrained MobileNetV2 backbone
base_model = MobileNetV2(
    input_shape=IMG_SIZE + (3,),
    include_top=False,  # Exclude the final classification layer
    weights='imagenet'  # Use pre-trained ImageNet weights
)

print("MobileNetV2 loaded successfully!")
print(f"Total layers in base model: {len(base_model.layers)}")
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h5
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 2s 0us/step
MobileNetV2 loaded successfully!
Total layers in base model: 154
```

Activity 3: Configure Transfer Learning Strategy

```
# Freeze all layers initially
for layer in base_model.layers:
    layer.trainable = False

# Unfreeze last N layers for fine-tuning
if N_LAST_LAYERS > 0:
    for layer in base_model.layers[-N_LAST_LAYERS:]:
        layer.trainable = True

# Count trainable parameters
trainable_count = sum([tf.size(w).numpy() for w in base_model.trainable_weights])
non_trainable_count = sum([tf.size(w).numpy() for w in base_model.non_trainable_weights])

print(f"Trainable parameters: {trainable_count:,}")
print(f"Non-trainable parameters: {non_trainable_count:,}")
print(f"Unfrozen last {N_LAST_LAYERS} layers for fine-tuning")
```

```
Trainable parameters: 732,480
Non-trainable parameters: 1,525,504
Unfrozen last 10 layers for fine-tuning
```

Activity 4: Build Complete Model Architecture

```
# Build the complete model
inputs = keras.Input(shape=IMG_SIZE + (3,))
x = base_model(inputs, training=False)  # Frozen BatchNorm layers
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dropout(0.35)(x)
x = layers.Dense(256, activation='relu')(x)
x = layers.Dropout(0.25)(x)
outputs = layers.Dense(NUM_CLASSES, activation='softmax')(x)

model = keras.Model(inputs, outputs, name="mobilenetv2_plant_disease_classifier")

# Display model summary
model.summary()
```

```
Model: "mobilenetv2_plant_disease_classifier"
```

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 256)	327,936
dropout_1 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 38)	9,766

```
Total params: 2,595,686 (9.90 MB)
Trainable params: 1,070,182 (4.08 MB)
Non-trainable params: 1,525,504 (5.82 MB)
```

Architecture Explanation:

1. Input Layer: Accepts 224x224x3 RGB images
2. MobileNetV2 Base: Pre-trained feature extractor (partially frozen)
3. GlobalAveragePooling2D: Reduces spatial dimensions to 1D
4. Dropout (0.35): Regularization to prevent overfitting
5. Dense (256): Fully connected layer with ReLU activation
6. Dropout (0.25): Additional regularization
7. Output Dense (38): Final classification layer with softmax

Milestone 3: Model Training

Activity 1: Compile the Model

```
# Compile model with appropriate settings
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-4),  # Low learning rate for fine-tuning
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

print("Model compiled successfully!")
print(f"Optimizer: Adam (lr=1e-4)")
print(f"Loss function: Categorical Crossentropy")
print(f"Metrics: Accuracy")
```

```
Model compiled successfully!
Optimizer: Adam (lr=1e-4)
Loss function: Categorical Crossentropy
Metrics: Accuracy
```

Compilation Settings Explanation:

- **Optimizer**: Adam with learning rate 1e-4 (low rate prevents disrupting pre-trained weights)
- **Loss Function**: Categorical crossentropy (standard for multi-class classification)
- **Metrics**: Accuracy for monitoring training progress

Activity 2: Setup Training Callbacks

```
# Define training callbacks
callbacks = [
    # Save best model based on validation accuracy
    keras.callbacks.ModelCheckpoint(
        "/content/mobilenetv2_best.keras",
        monitor='val_accuracy',
        save_best_only=True,
        verbose=1
    ),

    # Reduce learning rate when validation loss plateaus
    keras.callbacks.ReduceLROnPlateau(
        monitor='val_loss',
        factor=0.5,
        patience=3,
        verbose=1
    ),

    # Stop training if no improvement
    keras.callbacks.EarlyStopping(
        monitor='val_loss',
        patience=6,
        restore_best_weights=True,
        verbose=1
    )
]

print("Callbacks configured:")
print("  1. ModelCheckpoint - Saves best model")
print("  2. ReduceLROnPlateau - Adjusts learning rate")
print("  3. EarlyStopping - Prevents overfitting")
```

```
Callbacks configured:
  1. ModelCheckpoint - Saves best model
  2. ReduceLROnPlateau - Adjusts learning rate
  3. EarlyStopping - Prevents overfitting
```

Callbacks Explanation:

1. **ModelCheckpoint**: Automatically saves the model with the best validation accuracy
2. **ReduceLROnPlateau**: Reduces learning rate by 50% if validation loss doesn't improve for 3 epochs
3. **EarlyStopping**: Stops training if validation loss doesn't improve for 6 epochs

Activity 3: Train the Model

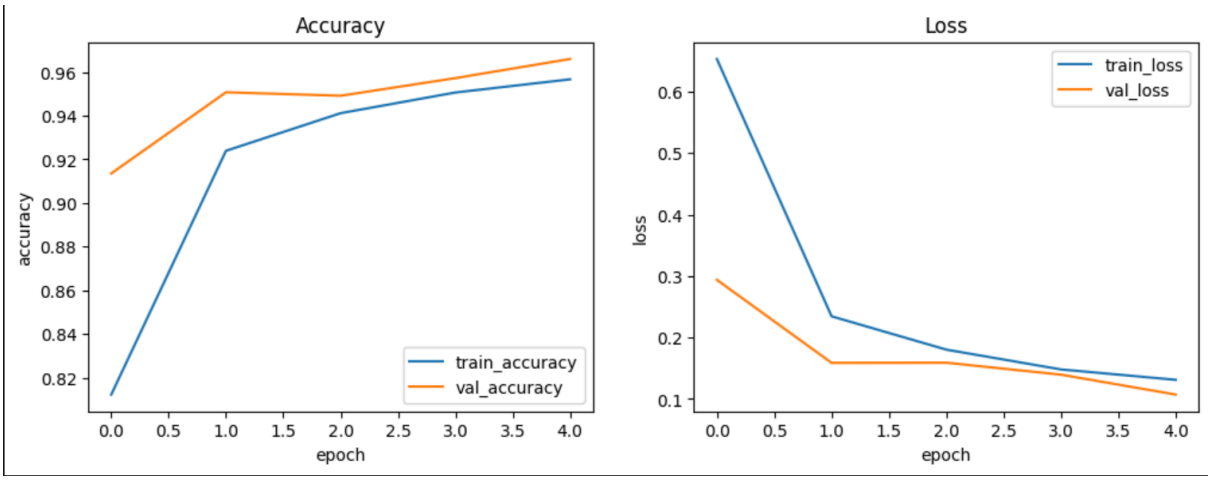
```
# ---------------- Train (single stage: some backbone layers trainable from start) ----------------
history = model.fit(
    train_gen,
    epochs=EPOCHS,
    validation_data=valid_gen,
    callbacks=callbacks
)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset` class should call `sup
  self._warn_if_super_not_called()
Epoch 1/5
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 0s 383ms/step - accuracy: 0.6634 - loss: 1.2439
Epoch 1: val_accuracy improved from -inf to 0.91356, saving model to /content/mobilenetv2_best.keras
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 905s 402ms/step - accuracy: 0.6635 - loss: 1.2436 - val_accuracy: 0.9136 - val_loss: 0.2938 - learning_rate: 1.0000e-04
Epoch 2/5
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 0s 377ms/step - accuracy: 0.9190 - loss: 0.2525
Epoch 2: val_accuracy improved from 0.91356 to 0.95077, saving model to /content/mobilenetv2_best.keras
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 854s 389ms/step - accuracy: 0.9190 - loss: 0.2524 - val_accuracy: 0.9508 - val_loss: 0.1587 - learning_rate: 1.0000e-04
Epoch 3/5
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 0s 373ms/step - accuracy: 0.9389 - loss: 0.1887
Epoch 3: val_accuracy did not improve from 0.95077
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 843s 384ms/step - accuracy: 0.9389 - loss: 0.1887 - val_accuracy: 0.9492 - val_loss: 0.1590 - learning_rate: 1.0000e-04
Epoch 4/5
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 0s 373ms/step - accuracy: 0.9496 - loss: 0.1503
Epoch 4: val_accuracy improved from 0.95077 to 0.95726, saving model to /content/mobilenetv2_best.keras
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 846s 385ms/step - accuracy: 0.9496 - loss: 0.1503 - val_accuracy: 0.9573 - val_loss: 0.1396 - learning_rate: 1.0000e-04
Epoch 5/5
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 0s 376ms/step - accuracy: 0.9566 - loss: 0.1325
Epoch 5: val_accuracy improved from 0.95726 to 0.96603, saving model to /content/mobilenetv2_best.keras
2197/2197 ━━━━━━━━━━━━━━━━━━━━ 851s 387ms/step - accuracy: 0.9566 - loss: 0.1325 - val_accuracy: 0.9660 - val_loss: 0.1072 - learning_rate: 1.0000e-04
Restoring model weights from the end of the best epoch: 5.
```

Activity 4: Visualize Training History

```
# ---------------- Plot training curves ----------------
plt.figure(figsize=(12,4))
plt.subplot(1,2,1)
plt.plot(history.history.get('accuracy',[]), label='train_accuracy')
plt.plot(history.history.get('val_accuracy',[]), label='val_accuracy')
plt.xlabel('epoch'); plt.ylabel('accuracy'); plt.legend(); plt.title('Accuracy')

plt.subplot(1,2,2)
plt.plot(history.history.get('loss',[]), label='train_loss')
plt.plot(history.history.get('val_loss',[]), label='val_loss')
plt.xlabel('epoch'); plt.ylabel('loss'); plt.legend(); plt.title('Loss')
plt.show()
```



Training Analysis:

The training curves show:

- Rapid initial learning: Training accuracy jumps from 66% to 92% in the first epoch
- Steady improvement: Validation accuracy improves from 91% to 96.6% over 5 epochs
- No overfitting: Training and validation curves remain close, indicating good generalization
- Convergence: Both loss curves decrease smoothly, showing stable training

Milestone 4: Model Evaluation and Testing

Activity 1: Evaluate Model Performance

```
# ---------------- Evaluate ----------------
val_loss, val_acc = model.evaluate(valid_gen)
print(f"Validation loss: {val_loss:.4f}, accuracy: {val_acc:.4f}")
```

```
550/550 ━━━━━━━━━━━━━━━━━━━━ 27s 49ms/step - accuracy: 0.9823 - loss: 0.0575
Validation loss: 0.1072, accuracy: 0.9660
```

Activity 2 : Save the Final Model

```
# ---------------- Save final model ----------------
final_path = "/content/mobilenetv2_final.keras"
model.save(final_path)
print("Saved final model to:", final_path)
```

```
Saved final model to: /content/mobilenetv2_final.keras
```

Milestone 5: Application Building

In this section, we will build a complete web application integrated with our trained MobileNetV2 model. The application provides an intuitive user interface where users can upload plant leaf images and receive instant disease detection results with actionable recommendations.

This milestone includes the following activities:

- Building HTML Templates
- Creating CSS Stylesheets
- Developing Flask Backend

Activity 1: Build Flask Backend (app.py)

File Location: `app.py` (in project root directory)

app.py 1 ×

app.py > ...

```python
@app.route('/predict', methods=['POST'])
def predict():
    if 'file' not in request.files:
        return jsonify({'error': 'No file uploaded'}), 400

    file = request.files['file']

    if file.filename == '':
        return jsonify({'error': 'No file selected'}), 400

    if file:
        filename = secure_filename(file.filename)
        filepath = os.path.join(app.config['UPLOAD_FOLDER'], filename)
        file.save(filepath)

        try:
            prediction = predict_image(filepath)

            # Save image to static folder for display
            static_filename = f"upload_{secrets.token_hex(8)}.jpg"
            static_path = os.path.join(app.config['STATIC_FOLDER'], 'images', static_filename)
            Image.open(filepath).save(static_path)

            # Store in session
            session['prediction'] = prediction
            session['image_path'] = f'images/{static_filename}'

            os.remove(filepath)  # Clean up temp file

            return jsonify({'success': True})
        except Exception as e:
            if os.path.exists(filepath):
                os.remove(filepath)
            return jsonify({'error': str(e)}), 500

@app.route('/result')
def result():
    prediction = session.get('prediction')
    image_path = session.get('image_path')

    if not prediction:
        return redirect(url_for('upload'))

    return render_template('result.html', prediction=prediction, image_path=image_path)

if __name__ == '__main__':
    load_model()
    app.run(debug=True, host='0.0.0.0', port=5000)
```

Run the code:

```
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production
WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.68.69:5000
Press CTRL+C to quit
 * Restarting with stat
Model loaded successfully from mobilenetv2_best.keras
 * Debugger is active!
 * Debugger PIN: 124-630-444
```

Activity 2: Building HTML Templates

For this project, we will create four HTML files that form the complete user interface. All HTML files should be saved in the templates folder.

Activity 2.1: Create Home Page (home.html)

The home page serves as the landing page for PlantCare AI, showcasing the application's features and capabilities.

File Location: templates/home.html

Key Features:

- Hero section with call-to-action
- Feature highlights (Instant Detection, High Accuracy, Easy to Use)
- Statistics display (38+ diseases, 95% accuracy, 10+ plant species)
- Responsive navigation bar

Code:

templates > dj about.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>About - Plant Disease Detection</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
    <!-- Navigation -->
    <nav class="navbar">
        <div class="nav-container">
            <div class="nav-logo">
                <div class="logo-icon">
                    <svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fi
                        <path d="M11 20A7 7 0 0 1 9.8 6.1C15.5 5 17 4.48 19 2c1 2 2 4.18 2 8 0 5.5-4.78 1
                        <path d="M2 21c0-3 1.85-5.36 5.08-6C9.5 14.52 12 13 13 12"></path>
                    </svg>
                </div>
                <span class="logo-text">PlantCare AI</span>
            </div>
            <ul class="nav-menu">
                <li><a href="/">Home</a></li>
                <li><a href="/about" class="active">About</a></li>
                <li><a href="/upload" class="btn-nav">Get Started</a></li>
            </ul>
        </div>
    </nav>

    <!-- About Hero -->
    <section class="about-hero">
        <div class="container">
            <h1 class="page-title">About Our Technology</h1>
            <p class="page-subtitle">Transforming agriculture with artificial intelligence</p>
        </div>
    </section>

    <!-- Mission Section -->
    <section class="mission-section">
        <div class="container">
```

SMARTBRIDGE
Let's Bridge the Gap

Smart Internz

PlantCare AI

Home About Get Started

AI-Powered Plant Disease Detection

Protect your crops with cutting-edge artificial intelligence. Instant disease identification for healthier plants and better yields.

Upload Image | Learn More

Why Choose PlantCare AI?

Instant Detection

Get results in seconds with our advanced AI model trained on thousands of plant images.

High Accuracy

Powered by MobileNetV2 architecture for precise disease identification across multiple plant species.

Easy to Use

Simple interface - just upload a photo and get instant diagnosis. No technical knowledge required.

38+	95%	10+	24/7
Plant Diseases	Accuracy Rate	Plant Species	Available

Ready to Protect Your Plants?

Start detecting plant diseases now with our AI-powered system

Get Started Now

About Plant Disease Detection System

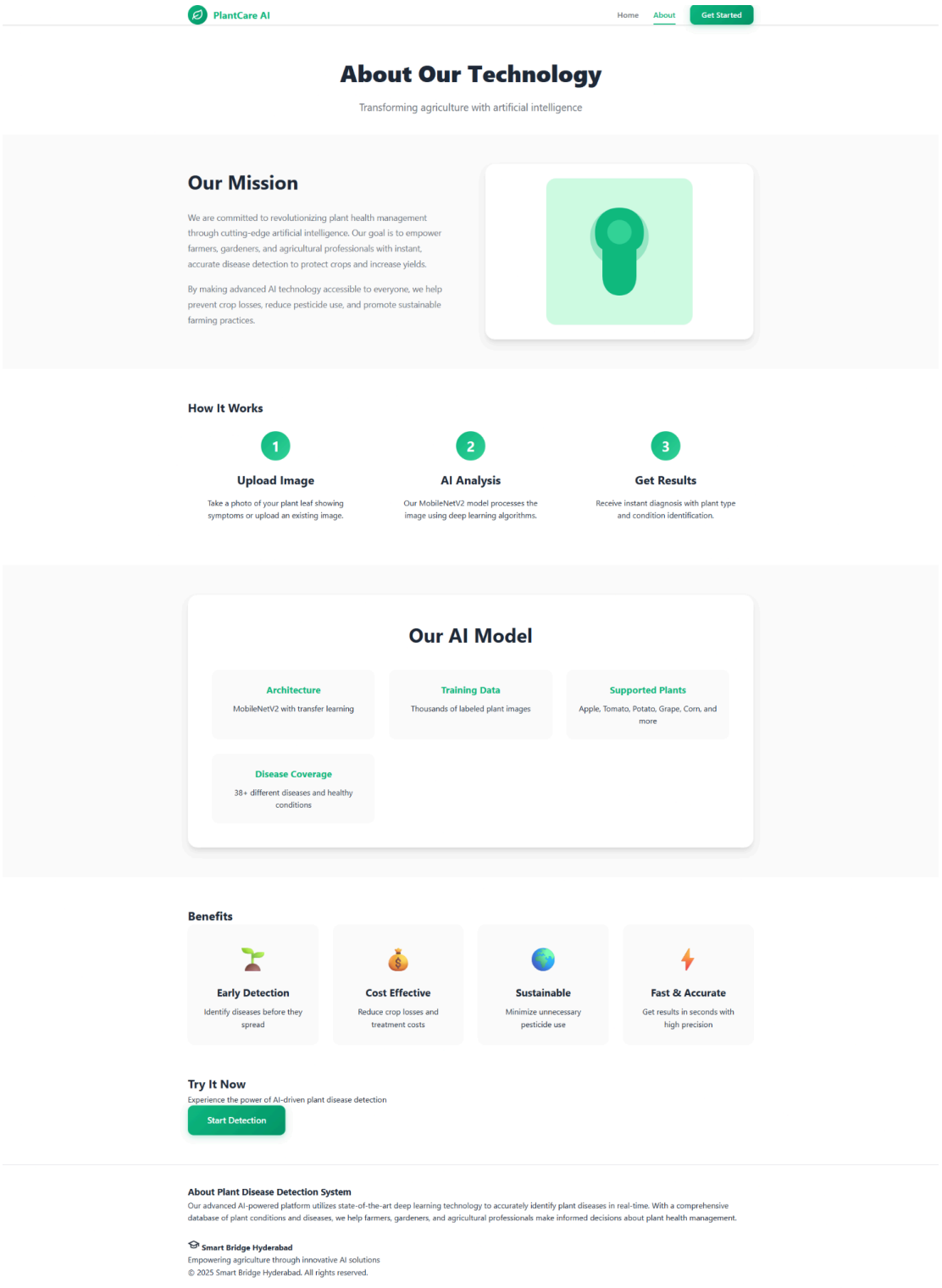
Our advanced AI-powered platform utilizes state-of-the-art deep learning technology to accurately identify plant diseases in real-time. With a comprehensive database of plant conditions and diseases, we help farmers, gardeners, and agricultural professionals make informed decisions about plant health management.

Smart Bridge Hyderabad
Empowering agriculture through innovative AI solutions

Activity 2.2: Create About Page (about.html)

The About page provides detailed information about the technology, mission, and AI model used in PlantCare AI.

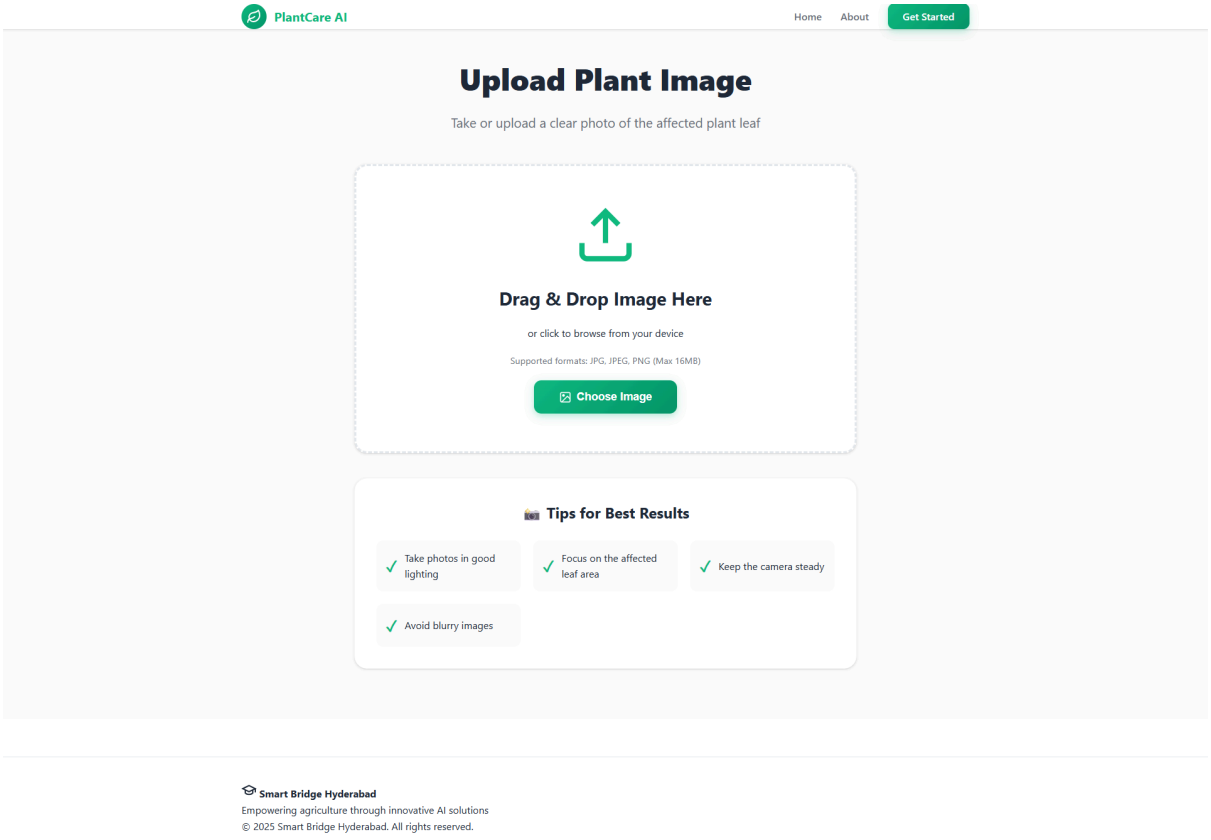
File Location: templates/about.html



Activity 2.3: Create Upload Page (upload.html)

The upload page allows users to select and upload plant leaf images for disease detection.

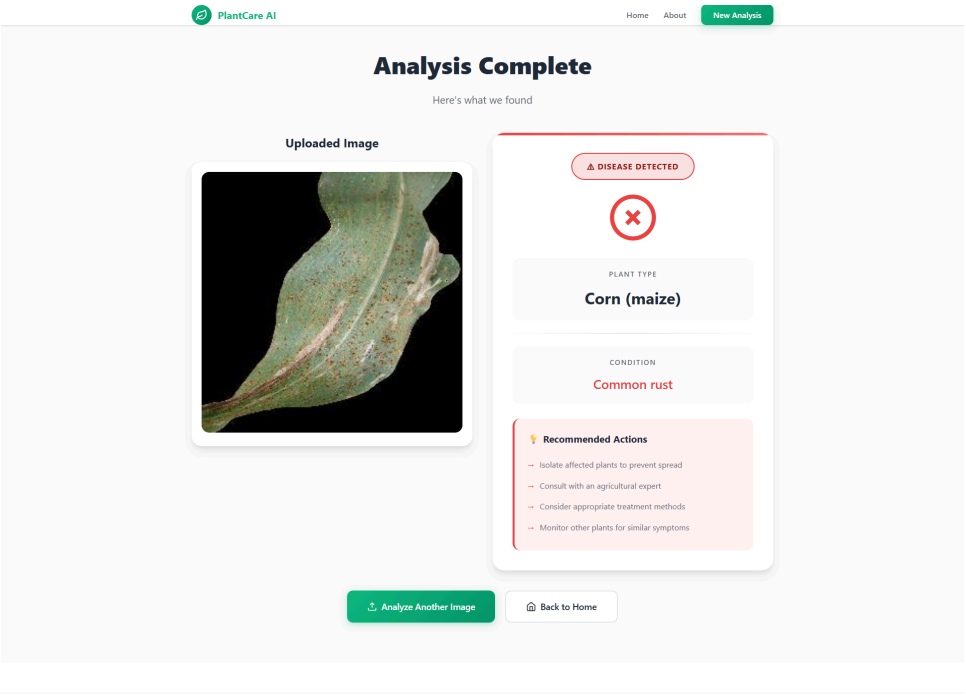
File Location: templates/upload.html

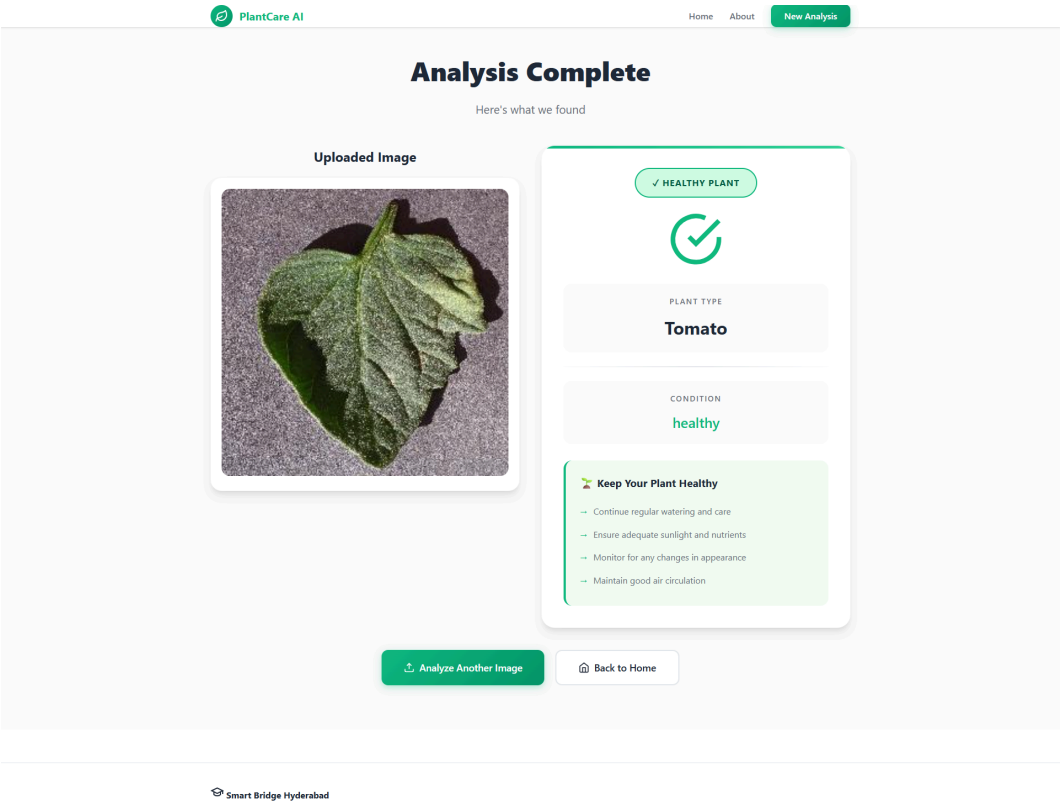


Activity 1.4: Create Result Page (result.html)

The result page displays the disease detection analysis with detailed information and recommendations.

File Location: templates/result.html





Future Implementations

Future plans for PlantCare AI focus on deployment and expansion:

Automated Agricultural Monitoring: Integrate the system into farm monitoring to enable real-time disease classification from captured images, automating reporting and intervention.

Mobile Application: Develop a user-friendly mobile app allowing home gardeners and small farmers to get instant disease analysis by simply taking a photo.

Educational Integration: Incorporate the model into agricultural training platforms for practical, hands-on learning in plant pathology.

Continuous Improvement: Expand the dataset to cover more diseases and plant species, and further fine-tune the MobileNetV2 architecture to ensure the highest possible accuracy across diverse environments.

Conclusion

The PlantCare AI project successfully developed an accurate and efficient AI-powered plant disease detection system. By leveraging the MobileNetV2 CNN through transfer learning, the model achieved a high validation accuracy of up to 96.6% using a dataset of over 87,000 images across 38 distinct disease classes.

The final product is a robust, full-stack web application (built with Flask) that seamlessly integrates the trained deep learning model into an intuitive user interface. This solution provides a reliable and scalable tool for precise and efficient disease identification, meeting the practical needs of farmers and gardeners.